

Diabetes Readmission Analysis

Predicting 30-Day Hospital Returns

DVA Capstone 2 — Final Project Report

SECTOR	Healthcare
TEAM ID	Section C, Group G-10
INSTITUTE	Newton School of Technology
GITHUB	github.com/tnshgarg/SectionC_G-10_DiabetesReadmissionAnalysis

TEAM MEMBER	ROLE
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2 Executive Summary

Hospital readmission within 30 days of discharge represents one of the most costly and preventable challenges in modern healthcare. The Diabetes Readmission Analysis is a data-driven initiative developed by Newton School of Technology's Section C, Group G-10. Drawing on a decade of clinical encounter records from 130 US hospitals, this project identifies the patient characteristics, treatment patterns, and hospital factors most strongly associated with early readmission, and translates them into actionable recommendations for hospital administrators, discharge planners, and clinical teams.

Business Challenge

Diabetic patients account for a disproportionate share of hospital readmissions. Beyond the clinical burden, each unplanned 30-day return triggers financial penalties under value-based care frameworks, strains bed capacity, and signals gaps in discharge planning or post-care support. Without data-driven visibility into which patients are at highest risk before discharge, clinical teams rely on intuition rather than evidence — resulting in avoidable returns and suboptimal resource allocation.

Analytical Approach

The team sourced the publicly available UCI Diabetes 130-US Hospitals Dataset, covering 101,766 clinical encounters from 1999 to 2008. A Python ETL pipeline standardised the raw data, engineered clinical features, and produced a fully clean analytical corpus of 101,763 records across 54 curated columns. Statistical analyses — including independent t-tests, Mann-Whitney U tests, chi-square tests, and Random Forest feature importance — validated key predictors. Findings were visualised through an interactive multi-view Tableau dashboard supporting both executive and operational decision-making.

Key Insights at a Glance

11.2%	4.4 Days	101,763	1999– 2008
30-Day Readmission Rate	Avg. Length of Stay	Clean Records Analysed	Data Time Span

- **Prior Inpatient Visits — The #1 Risk Factor:** Patients with 3+ prior inpatient visits in the preceding year are readmitted within 30 days at more than 2.5× the baseline rate. Historical utilisation is the strongest single predictor of early return.
- **Length of Stay Signals Complexity:** Readmitted patients stay an average of 0.8 days longer than those who are not readmitted — a statistically significant difference confirming that clinical complexity at the index stay directly elevates readmission risk.
- **Medication Change Reduces Risk:** Patients whose diabetic medication was actively adjusted during the encounter show lower 30-day readmission rates, indicating that proactive treatment optimisation before discharge is protective.

- **Diagnosis Type Drives Volume:** Circulatory and Diabetes primary diagnoses account for the highest absolute readmission volumes, making these patient groups the priority target for intervention programmes.
- **Emergency Admissions Most Vulnerable:** Patients admitted through the emergency pathway have significantly higher readmission rates than elective or urgent admissions — reflecting the severity and instability of their baseline condition.

Priority Recommendations

- Flag patients with 3+ prior inpatient visits for mandatory enhanced discharge planning and 7-day post-discharge follow-up calls.
- Implement a standardised medication reconciliation protocol before discharge for all patients with active medication changes.
- Prioritise intensive case management resources for Circulatory and Respiratory primary diagnosis groups.
- Launch a 30-day post-discharge remote monitoring pilot for high-risk patients identified by the compound risk profile.

3 Sector & Business Context

Overview of the Healthcare Industry

The US healthcare sector operates under increasing performance accountability through value-based care (VBC) models, including the Hospital Readmissions Reduction Programme (HRRP) administered by the Centers for Medicare & Medicaid Services (CMS). Under HRRP, hospitals with excess 30-day readmission rates for conditions including heart failure, pneumonia, and chronic obstructive pulmonary disease face direct financial penalties — up to 3% of total Medicare payments. Diabetic patients represent a particularly high-risk cohort: comorbid conditions, polypharmacy, and the chronic nature of diabetes management make readmission both clinically complex and financially significant for any hospital serving a substantial Medicare or Medicaid population.

Primary Decision-Makers

This analysis is designed to serve Hospital General Managers accountable for overall operational and financial performance, Discharge Planning Teams responsible for safe and timely patient transitions, Clinical Quality Directors overseeing care pathway adherence and patient outcomes, and Population Health Managers coordinating post-acute follow-up and community care resources.

Why 30-Day Readmissions Were Chosen as the Focus

The 30-day readmission window is the industry-standard regulatory boundary for preventable returns. Unlike longer-term outcomes, 30-day events are substantially within the hospital's sphere of influence — shaped by discharge preparation, medication management, follow-up scheduling, and patient education decisions made before the patient leaves the facility. Identifying the signatures of high-risk patients before discharge provides a direct, immediately actionable lever for clinical teams — without requiring capital investment in new facilities or staff.

Business Value of This Project

By enabling clinical and administrative teams to identify, segment, and prioritise high-risk diabetic patients before discharge, this project delivers value across three dimensions: financial protection through reduced CMS penalty exposure, operational efficiency through targeted allocation of discharge planning resources to the patients who need them most, and clinical quality improvement through evidence-based identification of the treatment factors most associated with stable post-discharge outcomes.

4 Problem Statement & Objectives

Formal Problem Definition

Which combinations of patient demographics, clinical encounter metrics, prior utilisation history, and treatment decisions — recorded at the point of the index hospital admission — are most predictive of 30-day readmission for diabetic patients? And what targeted clinical and operational interventions can hospital administrators implement to reduce preventable early returns without compromising discharge efficiency or patient throughput?

Project Scope

In Scope	Out of Scope
Historical clinical encounter data (1999–2008) from 130 US hospitals.	Real-time EHR or live PMS data integration.
Analysis of patient demographics, admission type, discharge disposition.	Post-2020 healthcare behavioural shifts or pandemic effects.
Medication change, insulin management, and diagnosis group patterns.	Individual physician or nurse-level performance analysis.
Prior visit history (inpatient, emergency, outpatient) as risk predictors.	Cost and reimbursement data or insurance billing outcomes.
Interactive Tableau dashboard for executive and operational audiences.	Competitor hospital benchmarking or regional market comparisons.

Success Criteria

- Delivery of a minimum of eight statistically supported, business-relevant insights in clinical decision language.
- A functional two-view Tableau dashboard accessible via Tableau Public, covering executive summary and operational drill-down.
- Concrete, prioritised recommendations with clearly articulated expected clinical and financial impact.
- A reproducible Python ETL pipeline committed to the team GitHub repository with zero null values in the output dataset.

5 Data Description

Source & Access

The dataset used is the publicly available Diabetes 130-US Hospitals for Years 1999–2008 dataset, originally compiled and published on the UCI Machine Learning Repository. The raw file (diabetic_data.csv) was accessed directly via the UCI platform and committed to the project GitHub repository for reproducibility and auditability.

Data Structure

101,766	101,763	50	54
Raw Rows	Processed Rows	Original Columns	Final Columns

Key Fields

Field	Description
Readmitted Under 30 Days	PRIMARY TARGET — Binary flag: 1 if patient readmitted within 30 days of discharge.
Days in Hospital	Number of days between admission and discharge. KPI 2 (Average Length of Stay).
Age	Age interval converted to numerical midpoints (5–95). Used for demographic segmentation.
Prior Inpatient Visits	Number of inpatient visits in the preceding year — identified as the strongest single predictor.
Prior Emergency Visits	Emergency department visits in the preceding year — compound risk indicator.
Admission Type	How the patient was admitted: Emergency, Urgent, Elective, Trauma Center, etc.
Discharge Disposition	Destination after discharge: Home, SNF Transfer, Hospice, etc.
Primary Diagnosis Group	ICD9 codes mapped to 17 clinical groups (Circulatory, Diabetes, Respiratory, etc.).
Medication Changed	Whether any diabetic medication dosage was adjusted during the encounter.
Insulin	Insulin prescription status: No, Steady, Up (increased), Down (decreased).
A1C Result	A1c test result: Norm, >7, >8, or Not Tested — glycaemic control indicator.
Race / Gender	Demographic attributes for disparity and segmentation analysis.

Data Limitations & Known Biases

- The dataset covers 1999–2008 and does not reflect post-ACA healthcare reforms, modern insulin analogue adoption, or COVID-era telehealth shifts.

- Weight data was missing for the vast majority of records and could not be used analytically, removing a clinically significant predictor.
- Medical specialty had high missingness; imputing as 'Unknown' masks potentially meaningful physician-level variation in care protocols.
- The dataset is drawn from US hospitals only and may not generalise to healthcare systems with different readmission incentive structures.
- Race and socioeconomic data are limited — disparities may be understated due to imputation choices for missing race values.

6 Data Cleaning & ETL Pipeline

All data extraction, transformation, and loading (ETL) operations were executed exclusively in Python, in compliance with the DVA Capstone 2 requirements. The pipeline is documented across two committed artefacts: `scripts/etl_pipeline.py` (the production-grade script) and `notebooks/02_cleaning.ipynb` (the annotated, step-by-step cleaning notebook).

Missing Value Treatment

Column / Issue	Action Taken	Rationale
Raw '?' placeholder (widespread)	Converted to NaN, then imputed with explicit labels	Raw data used '?' as missing indicator — must be standardised
weight — high missingness	Imputed as 'Not Recorded'	Dropping rows would lose >97% of dataset; label preserves missingness signal
race — 2,273 null records	Imputed as 'Unknown'	Demographic integrity maintained; unknown is analytically honest
payer_code / medical_specialty	Imputed as 'Unknown'	Preserves rows while flagging missing clinical metadata
max_glu_serum / A1Cresult	Imputed as 'Not Tested'	Distinguishes untested from abnormal — clinically meaningful distinction
gender — 3 'Unknown/Invalid' records	Rows removed entirely	Negligible volume; invalid demographic records excluded
examide / citoglipton (constant columns)	Columns dropped entirely	Both contained only one value ('No') — zero analytical variance

Data Standardisation & Feature Engineering

Derived Feature / Transformation	Method	Business Purpose
Age numerical midpoints	Interval '[50-60]' → midpoint 55	Enables statistical correlation of age with readmission probability
Admission Type labels	Numeric IDs mapped to clinical labels (Emergency, Urgent, Elective...)	Human-readable values for dashboard filters and clinical interpretation
Discharge Disposition labels	29 numeric codes mapped to 15 clinical disposition labels	Enables analysis of discharge destination as a readmission risk factor
Primary / Secondary / Additional Diagnosis Group	ICD9 codes classified into 17 high-level clinical groups	Transforms raw codes into interpretable categories (Circulatory, Diabetes, etc.)
Readmitted Under 30 Days	readmitted == '<30' → binary 1/0	Primary KPI target variable for all risk modelling and segmentation
Readmitted After 30 Days	readmitted == '>30' → binary 1/0	Supports differentiation of early vs late readmission patterns
Not Readmitted	readmitted == 'NO' → binary 1/0	Baseline comparison group for all statistical tests
Column renaming to Tableau-friendly labels	All snake_case renamed to title-case professional labels	Direct compatibility with Tableau field references; no custom aliases required

7 KPI & Metric Framework

Two core KPIs were defined to align analytical outputs with the project's business objectives. Each metric was computed within the Python notebooks and visualised as a headline tile in the Tableau dashboard.

KPI 1 — 30-Day Readmission Rate (%)

Formula	$\text{Count(Readmitted Under 30 Days = 1)} / \text{Total Admissions} \times 100$
Computed Value	$\approx 11.2\%$ of all encounters resulted in a readmission within 30 days
Business Relevance	The primary regulatory KPI under CMS HRRP. A rising rate indicates deteriorating discharge quality or inadequate post-care support. Drives penalty exposure directly.

KPI 2 — Average Length of Stay (Days)

Formula	$\text{Average(Days in Hospital) across all encounters}$
Computed Value	≈ 4.4 days overall; readmitted patients average ~ 0.8 days longer than non-readmitted
Business Relevance	Longer stays indicate clinical complexity. The gap between readmitted and non-readmitted cohorts confirms length of stay as a significant risk stratification signal.

8 Exploratory Data Analysis

Exploratory Data Analysis was conducted across five analytical dimensions — target variable distribution, demographic segmentation, admission and discharge patterns, clinical encounter metrics, and medication / treatment impact — to build a comprehensive picture of the readmission landscape. All charts and outputs are committed to notebooks/03_eda.ipynb.

Target Variable Distribution — Readmission Profile

The dataset reveals that approximately 11.2% of encounters result in readmission within 30 days, 35.2% result in readmission after 30 days, and the remaining 53.6% are never readmitted. The 30-day cohort, while numerically the smallest, represents the highest-stakes group from a regulatory and clinical standpoint, and is therefore the sole focus of all downstream analyses.

Demographic Analysis — Age, Race & Gender

Age analysis reveals a non-linear relationship with readmission: patients in the 70–80 age bracket (midpoint 75) exhibit the highest absolute encounter volume and elevated readmission rates, reflecting the comorbidity burden common in elderly diabetic patients. Younger age groups (under 45) show lower readmission rates, though sample sizes are smaller. Race analysis identifies African American patients as having measurably higher readmission rates than Caucasian or Hispanic patients — a disparity warranting targeted equity-focused interventions. Gender differences in readmission rates are statistically present but modest in magnitude.

Admission & Discharge Pattern Analysis

Emergency admissions account for the largest patient volume and significantly higher 30-day readmission rates compared to elective or urgent admissions. This is expected: emergency pathways capture the most clinically unstable patients with the least preparation time. Discharge disposition is an equally powerful differentiator — patients discharged to home without follow-up support show materially higher readmission rates than those transitioning to skilled nursing facilities (SNF) or receiving home health services. This suggests that structured post-discharge care coordination is a meaningful protective factor.

Clinical Feature Analysis — Prior Visit History

Prior inpatient visit count is the single most powerful predictor identified in the EDA. The relationship is steeply positive: patients with zero prior inpatient visits in the preceding year have a readmission rate of approximately 9%, while those with 3 or more visits exceed 25%. This creates an immediate, objective risk stratification criterion that clinical teams can apply without any model: high prior visit count = high readmission risk, requiring enhanced discharge planning by default. Prior emergency visit count shows a similar, though slightly less steep, positive relationship.

Medication & Treatment Impact Analysis

Medication change analysis reveals a consistent and clinically meaningful pattern: patients whose diabetic medication was actively adjusted during the encounter have lower 30-day readmission rates than those whose medication remained unchanged. This counterintuitive finding suggests that medication changes reflect clinical engagement — a care team that actively optimises treatment before discharge produces more stable post-discharge outcomes. Insulin status analysis further confirms that dosage management protocol matters: patients with no prescribed insulin and those with stable insulin doses show the lowest readmission rates, while those with down-titrated insulin show elevated rates — potentially reflecting clinical deterioration at the time of discharge.

Correlation Summary

Variable Pair	Relationship	Business Implication
Prior Inpatient Visits × Readmit <30d	Strong Positive	Strongest predictor — flag patients with 3+ prior visits for enhanced discharge planning
Days in Hospital × Readmit <30d	Positive	Longer stays signal complexity that persists post-discharge
Emergency Admission × Readmit <30d	Positive	Highest-risk entry pathway; dedicated discharge resources required
Medication Changed × Readmit <30d	Negative	Active treatment optimisation is protective; stable meds without change may indicate under-treatment
African American Race × Readmit <30d	Positive	Equity gap — targeted post-discharge support programmes warranted
Discharge to SNF/Home Health × Readmit <30d	Negative	Structured post-acute care reduces early return probability

9 Statistical Analysis

Statistical methods were applied to move beyond descriptive observations and validate findings with inferential rigour. All analyses are committed to notebooks/04_statistical_analysis.ipynb. Methods applied include independent samples t-tests, Mann-Whitney U tests, chi-square tests of independence, and Random Forest feature importance scoring.

Test	Hypothesis	Key Metric	Significance	Business Story
Independent T-Test	Length of Stay differs between readmitted and non-readmitted patients	Readmitted: +0.8 days longer	$p < 0.001$ — High	Longer stays = higher complexity = elevated return risk
Mann-Whitney U Test	Prior inpatient visits higher for readmitted patients	Readmitted cohort: 2.5× higher prior visits	$p < 0.0001$ — Critical	Historical utilisation is the dominant readmission predictor
Chi-Square Test	Medication change status is associated with readmission	Chi ² statistically significant	$p < 0.001$ — High	Active treatment changes are protective; unchanged meds may signal gaps
Chi-Square Test	Insulin dosage management is associated with readmission	Dosage adjustments correlate with outcomes	$p < 0.001$ — Medium	Insulin protocol is a key lever in post-discharge stability
Chi-Square Test	Primary diagnosis group is independent of readmission status	Circulatory & Diabetes drive highest volume	$p < 0.0001$ — Critical	Diagnosis category is a major determinant of readmission burden
Random Forest Feature Importance	Which numerical features best predict 30-day readmission?	Top: Prior Inpatient Visits, Days in Hospital, Lab Procedures	Model AUC > 0.70	Feature importance confirms prior utilisation as the dominant signal

High-Risk Patient Profile — Compound Risk Analysis

A compound risk analysis identified the highest-risk patient profile as the combination of:

HIGH-RISK PROFILE
Emergency Admission + 3+ Prior Inpatient Visits + Circulatory/Diabetes Diagnosis + No Post-Discharge Care Plan

Patients matching all four criteria exhibit 30-day readmission rates well above the dataset average of 11.2%, representing the highest-priority target population for proactive clinical intervention. This profile should trigger automatic enhanced discharge planning protocols, mandatory medication reconciliation, and 7-day post-discharge outreach by the care coordination team.

10 Tableau Dashboard Design

Two interactive Tableau dashboards were designed and published to Tableau Public to provide hospital administrators and clinical teams with analytical access to the project's core findings. The dashboards are structured into an Executive Summary View and a Patient Risk Analysis View, serving different decision-making timeframes and user personas.

Dashboard 1 — Executive Overview

The Executive Overview dashboard provides a high-level summary of readmission performance across the full dataset period. Key headline metrics — overall 30-day readmission rate, average length of stay, total encounters, and readmission distribution — are displayed as large-format KPI tiles. Trend charts surface readmission volume by age group and demographic breakdown, enabling executive audiences to identify the highest-impact patient segments at a glance without requiring drill-down interaction.

Dashboard 2 — Patient Risk Analysis

The Patient Risk Analysis dashboard enables clinical teams to interrogate the data at a granular level, exploring readmission rates broken down by prior inpatient visit count, medication change status, insulin dosage, primary diagnosis group, and admission type. Side-by-side comparisons between readmitted and non-readmitted cohorts across clinical metrics allow discharge planners to immediately identify which patient subgroups require enhanced intervention before discharge.

Dashboard 3 — Treatment & Outcomes

The Treatment & Outcomes dashboard focuses specifically on the relationship between medication management decisions — medication changes, insulin protocol, A1C result — and 30-day readmission outcomes. This dashboard is designed for clinical quality directors evaluating whether pharmacological interventions at the index stay are adequately protecting against early return.

Interactive Filters

- **Admission Type:** Toggle between Emergency, Urgent, Elective, and Trauma Center admissions.
- **Age Group:** Filter by patient age bracket to identify generational readmission patterns.
- **Race / Gender:** Demographic filters for equity analysis and disparity identification.
- **Primary Diagnosis Group:** Isolate readmission patterns for Circulatory, Diabetes, Respiratory, and other clinical categories.
- **Medication Changed:** Compare outcomes for patients whose medication was adjusted vs. unchanged.
- **Discharge Disposition:** Explore how post-discharge destination type affects readmission probability.

Dashboard URL:

https://public.tableau.com/app/profile/tanish.garg1945/viz/DVACapstoneDiabetesReadmissionAnalysis/Dashboard_Patient_Risk_Analysis

11 Insights Summary

The following eight insights represent the most decision-relevant findings from the combined EDA and statistical analyses. Each is framed in operational language to directly inform clinical and administrative action.

1

Prior Inpatient Visits — The Dominant Predictor

Patients with 3 or more prior inpatient visits in the preceding year have a 30-day readmission rate exceeding 25% — more than 2.5× the overall average. This metric is immediately available at the time of the index admission, making it the most actionable risk flag available to discharge planners without any model or scoring system.

2

Length of Stay Signals Unresolved Clinical Complexity

Patients readmitted within 30 days stayed an average of 0.8 days longer during their index admission than those who were not readmitted ($p < 0.001$). Extended stays indicate clinical instability that persists beyond discharge — and these patients require proactive follow-up scheduling before they leave the facility.

3

Emergency Admissions Carry the Highest Readmission Burden

Emergency admission pathway is associated with significantly elevated 30-day readmission rates versus elective or urgent pathways. Emergency patients arrive with the least preparation and the highest clinical instability — making targeted discharge planning for this cohort the single largest volume opportunity for readmission reduction.

4

Medication Change is Protective — Active Treatment Optimisation Matters

Patients whose diabetic medication was adjusted during the encounter exhibit lower readmission rates than those with no changes. Proactive medication reconciliation and dose optimisation before discharge appear to stabilise glycaemic control and reduce early return. Unchanged medication in a complex diabetic patient may warrant clinical review before discharge is authorised.

5

Circulatory and Diabetes Diagnoses Drive the Majority of Readmission Volume

Patients with Circulatory or Diabetes-primary diagnoses account for the largest absolute volumes of 30-day readmissions. Condition-specific discharge protocols — with standardised medication checklists, follow-up appointments, and patient education — for these two groups alone would address the majority of preventable returns.

6

African American Patients Face a Measurable Readmission Equity Gap

African American patients exhibit a materially higher 30-day readmission rate than Caucasian or Hispanic patients after controlling for age. This disparity is consistent with nationally documented inequities in chronic disease management and post-discharge support access. Equity-specific outreach programmes — including community health worker follow-up — are warranted for this population.

7

Discharge to Home Without Follow-up Elevates Risk

Patients discharged directly home without structured follow-up support (home health services or SNF transition) show higher readmission rates than those with formal post-acute care arrangements. Ensuring every high-risk patient has a confirmed 7-day follow-up appointment before discharge is a low-cost, high-impact protective intervention.

8

A1C Testing Gap — Untested Patients at Higher Risk

A meaningful proportion of encounters record A1C as 'Not Tested', and this group exhibits elevated readmission rates compared to patients with tested results — even abnormal ones. Failure to test A1C at admission may reflect either low clinical acuity screening or resource constraints. Implementing a universal A1C testing protocol for all diabetic admissions is likely to improve both discharge planning quality and post-acute glycaemic management.

12 Recommendations

The following four recommendations are prioritised by expected clinical impact and implementation feasibility. Each is directly traceable to a specific analytical finding.

#1 Mandatory Enhanced Discharge Planning for High Prior-Visit Patients	
INSIGHT	Patients with 3+ prior inpatient visits in the preceding year have a 30-day readmission rate exceeding 2.5× the baseline.
ACTION	Implement an automatic EHR flag that triggers mandatory enhanced discharge planning — including a confirmed 7-day post-discharge follow-up appointment, medication reconciliation review, and care coordinator assignment — for every patient with 3+ prior inpatient visits at the time of index admission. This requires no model or scoring tool, only an EHR rule that is implementable within days.
IMPACT	Directly targets the highest-risk identifiable patient segment before discharge. Feasible within 1–2 weeks via EHR configuration. Expected to reduce readmission rates in this cohort by 15–25% based on published enhanced discharge planning literature.

#2 Standardised Medication Reconciliation Protocol Before Discharge	
INSIGHT	Active medication changes during the encounter are associated with lower readmission rates — indicating that proactive treatment optimisation is protective. Patients with unchanged complex medication regimens may be discharged without adequate optimisation.
ACTION	Mandate a structured medication reconciliation checklist — reviewed by a clinical pharmacist or attending physician — for all diabetic patients before discharge authorisation. The checklist should specifically flag unchanged medication in patients with poor glycaemic control (A1C > 8 or 'Not Tested') for attending review. Pair with patient-facing medication education at discharge.
IMPACT	Addresses the direct pharmacological driver of readmission in a population where medication management is central to outcomes. Pharmacy-driven reconciliation programmes have demonstrated 10–20% readmission reductions in diabetic cohorts in peer-reviewed studies.

#3 Condition-Specific Discharge Protocols for Circulatory & Diabetes Diagnoses	
INSIGHT	Circulatory and Diabetes primary diagnosis groups account for the majority of absolute 30-day readmission volume — making these two condition categories the highest-leverage target for protocol-based intervention.
ACTION	Develop and deploy standardised, condition-specific discharge checklists for Circulatory and Diabetes-primary patients, covering: (a) confirmed specialist follow-up within 7 days, (b) written patient education materials on warning signs requiring immediate return, (c) dietitian consult before discharge, and (d) social work referral for patients without confirmed home support. Pilot in high-volume units before hospital-wide rollout.
IMPACT	Addresses the largest absolute volume of preventable readmissions. Protocol-driven discharge programmes consistently demonstrate 8–18% readmission reductions for cardiac and diabetic cohorts. Implementation cost is primarily staff time for checklist development and workflow integration.

#4 Equity-Targeted Post-Discharge Outreach for African American Patients	
INSIGHT	African American patients exhibit a measurably higher 30-day readmission rate than other demographic groups — a disparity documented nationally and confirmed in this

	dataset.
ACTION	Establish a targeted post-discharge outreach programme — delivered by community health workers or trained care navigators — for African American diabetic patients discharged to home without formal post-acute support. Outreach should include a 48-hour post-discharge check-in call, barriers-to-care assessment, and referral to community diabetes management resources where available. Track readmission rates for this cohort as a standalone equity KPI.
IMPACT	Reduces the documented equity gap in readmission outcomes. Community health worker programmes have demonstrated 20–30% reductions in readmissions for high-risk minority populations in peer-reviewed interventional studies. Strengthens the hospital's value-based care equity reporting and positions it favourably for CMS quality benchmarks.

13 Impact Estimation

While precise financial projections require institution-specific cost data beyond the scope of this dataset, the following estimates are directionally grounded in the analytical findings and published healthcare literature.

Clinical Impact — Readmission Reduction

The dataset contains approximately 11,400 encounters resulting in 30-day readmission. If Recommendations 1 and 2 together achieve a conservative 15% reduction in the high-risk patient cohort (approximately the top 30% of patients by prior visit count), this would represent roughly 500–700 prevented readmissions annually for a 300-bed hospital running at comparable volumes. Each prevented readmission eliminates both the direct clinical cost and the CMS penalty exposure associated with excess readmission rates.

Financial Impact — CMS Penalty Avoidance

Under the Hospital Readmissions Reduction Programme, a hospital with excess diabetic readmission rates faces penalties of up to 3% of total Medicare payments. For a mid-sized hospital receiving \$50M annually in Medicare reimbursements, moving from the penalty threshold back to the baseline readmission rate through the interventions described could protect up to \$1.5M per year in reimbursement. The implementation cost of the recommended interventions — primarily staff time for protocol development and EHR configuration — is typically recoverable within the first operating quarter.

Urgency — Why Act Now?

CMS has steadily expanded the list of conditions subject to HRRP penalties, and value-based care transition continues to accelerate. Hospitals that delay implementing systematic readmission risk identification will find their penalty exposure growing with each reporting period. The interventions identified in this project are implementable within weeks using existing EHR infrastructure — the opportunity cost of inaction compounds with every discharge cycle.

14 Limitations

Limitation	Detail
Temporal Data Gap	The dataset covers 1999–2008. Post-ACA readmission incentive structures, modern GLP-1 receptor agonist adoption, and telehealth-driven post-discharge monitoring are not captured. Conclusions should be re-validated against current EHR data before policy implementation.
Missing Weight Data	Weight data was missing for the vast majority of records. Body mass index (BMI) is a clinically significant predictor of diabetic comorbidity and readmission — its absence limits the predictive completeness of this analysis.
Institutional Scope	The 130 hospitals included are US institutions. Readmission patterns, incentive structures, and care protocols in other healthcare systems (NHS, universal coverage models) may differ substantially.
CLV / Patient Lifetime Value Absent	Without multi-encounter patient tracking across institutions, it is not possible to compute true patient-lifetime readmission trajectories. The 30-day flag is per-encounter, not per-patient lifetime.
Imputation Assumptions	Imputing race as 'Unknown' and medical specialty as 'Unknown' masks potentially meaningful variation. A propensity-score or multiple-imputation approach would yield more nuanced estimates of these variables' true impact.
Socioeconomic Data Absent	Insurance payer code was retained but imputed for missingness. True socioeconomic status, housing stability, and food security — known social determinants of readmission — are not available in this dataset.

15 Future Scope

Machine Learning — Readmission Probability Scoring

The binary classification structure of the Readmitted Under 30 Days target variable makes this dataset an ideal candidate for supervised ML modelling. A gradient boosting classifier (XGBoost or LightGBM) trained on the engineered feature set could generate a real-time Readmission Risk Score at the point of discharge authorisation. This score could be embedded directly in the EHR to trigger automated clinical decision support — for example, mandatory care coordinator review for any patient scoring above a defined risk threshold — without requiring manual clinical judgement for each case.

Social Determinants of Health Integration

Incorporating social determinants of health (SDOH) data — housing stability, food access, transportation availability, and social support networks — would substantially improve predictive model accuracy. CMS has increasingly mandated SDOH data collection through ICD-10-CM Z-codes; integrating these into the analytical framework would enable a more complete picture of post-discharge risk.

Real-Time EHR Integration & Live Dashboard

Connecting the Tableau dashboard to a live EHR data feed via an API or scheduled ETL pipeline would enable true real-time operational monitoring. Discharge planners could view intraday high-risk patient lists, track readmission rates against targets, and receive automated alerts when high-risk discharges are being processed without an enhanced discharge planning record. This evolution from historical analysis to live clinical decision support represents the highest-value application of the analytical framework developed in this project.

16 Conclusion

The Diabetes Readmission Analysis project demonstrates that rigorous analysis of historical clinical encounter data can yield commercially significant, immediately actionable intelligence — even from a dataset spanning a decade. Through a structured Python ETL pipeline and a multi-view Tableau dashboard, the team surfaced a clear and consistent finding: prior inpatient visit history, clinical complexity at the index stay, and the adequacy of discharge preparation are the primary structural drivers of 30-day readmission risk in diabetic patients.

The data confirms that a small subset of patients — identifiable before discharge using a single objective criterion (3+ prior inpatient visits) — accounts for a disproportionate share of preventable returns. The business case for action is compelling: the interventions required are low-cost and operationally straightforward, the financial upside through CMS penalty avoidance is material, and the clinical benefit in terms of patient outcomes and equity is both measurable and immediately deliverable.

By investing in data-driven discharge planning today — beginning with prior visit flagging and structured medication reconciliation — hospital administrators can convert analytical insight into prevented readmissions, protected revenue, and a more equitable and resilient care model for the patients who depend on it most.

17 Appendix

Artefact	Path	Description
Data Dictionary	docs/data_dictionary.md	Full column-level definitions for all 50 original and 54 processed fields.
ETL Pipeline	scripts/etl_pipeline.py	Production-grade Python script for end-to-end data extraction, cleaning, and feature engineering.
Extraction Notebook	notebooks/01_extraction.ipynb	Dataset loading, initial inspection, and null/duplicate profiling.
Cleaning Notebook	notebooks/02_cleaning.ipynb	Annotated step-by-step cleaning log with transformation rationale.
EDA Notebook	notebooks/03_eda.ipynb	Full exploratory analysis with demographic, clinical, and treatment charts.
Statistical Analysis	notebooks/04_statistical_analysis.ipynb	Hypothesis testing results, feature importance, and compound risk profile.
Final Load Prep	notebooks/05_final_load_prep.ipynb	Final dataset validation and Tableau-ready export preparation.
Tableau Screenshots	tableau/screenshots/	PNG exports of all three dashboard views including filter states.
Dashboard Links	tableau/dashboard_links.md	Tableau Public URLs for all published dashboards.
Processed Data	data/processed/cleaned_diabetic_data.csv	Final clean analytical dataset (101,763 rows × 54 columns, zero null values).

18 Contribution Matrix

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Dhanvin Vadlamudi	Support	Support	Support	Support	Support	Owner	Owner
Ayush Mittal	Owner	Support	Support	Support	Support	Support	Support
Gargi Srivastava	Support	Owner	Support	Support	Support	Support	Support
Tanish Garg	Support	Support	Owner	Owner	Support	Support	Support
Sumit Yadav	Support	Support	Support	Support	Owner	Support	Support
Lakshya	Support	Support	Support	Support	Support	Support	Owner

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, pull request history, and submitted project artefacts.

Team Lead Signature: ____Dhanvin Vadlamudi____ **Date:** ____28th April 2026____